STEP -1

```
Loaded 10 crops from the corpus folder.

========== FARM ASSIST ==========
1. Search (crop / disease / pest / anything)
2. List all crops
3. Compare string-search algorithms (Naive/KMP/Z/Rabin-Karp)
4. Water Distribution (Max Flow)
5. Crop -> Field Assignment (Bipartite Matching)
6. Field Visit Route (Bitmask DP / TSP)
7. Field Inspection Planner (Vertex Cover Approximation)
8. Crops of the Day (Reservoir Sampling)
9. Exit
Enter your choice: 2
1. Rice
2. Wheat
3. Maize
4. Cotton
5. Groundnut
6. Sugarcane
7. Tomato
8. Chilli
9. Turmeric
10. Soybean
```

STEP-2

```
Enter your choice: 1
Enter crop / disease / keyword: rice

========== SEARCH RESULTS ==========

========== CROP DETAILS ==========
Crop Name: Rice
Season: Kharif
Soil: Clayey loam
Diseases: Blast, Bacterial leaf blight
Pests: Stem borer, Brown planthopper
Fertilizer: Urea, DAP
Water Requirement: 1200 mm
==================================
```

STEP -3

```
Enter your choice: 1
Enter crop / disease / keyword: leaf curl

========== SEARCH RESULTS ==========
"leaf curl" found in 3 crop(s):

 1) Cotton
      - Diseases: Wilt, Leaf curl

 2) Tomato
      - Diseases: Early blight, Leaf curl

 3) Chilli
      - Diseases: Anthracnose, Leaf curl

Enter result number for full crop details (0 to go back): 1

========== CROP DETAILS ==========
Crop Name: Cotton
Season: Kharif
Soil: Black soil
Diseases: Wilt, Leaf curl
Pests: Bollworm, Whitefly
Fertilizer: Urea, MOP
Water Requirement: 800 mm
==================================
```

STEP-4

```
Enter your choice: 3
Enter pattern to benchmark: urea

========== STRING SEARCH BENCHMARK ==========
Corpus size: 2241 chars, pattern: "urea"

Algorithm     Matches   Time (us)   Comparisons  Complexity
Naive         6         330         2285         O(n*m)
KMP           6         125         2266         O(n+m)
Z-Algorithm   6         2866        -            O(n+m)
Rabin-Karp    6         503         -            O(n+m) avg

All four must report the same match count (correctness check).
```

```
Enter your choice: 1
Enter crop / disease / keyword: whete

========== SEARCH RESULTS ==========
No exact match for "whete". Did you mean: "Wheat"?

========== CROP DETAILS ==========
Crop Name: Wheat
Season: Rabi
Soil: Loam
Diseases: Rust, Powdery mildew
Pests: Aphids, Termites
Fertilizer: Urea, Potash
Water Requirement: 450 mm
==================================
```

STEP -5

```
Enter your choice: 4

========== WATER DISTRIBUTION (Max Flow) ==========
Number of locations (numbered 0..n-1): 6
Source location (e.g. reservoir): 0
Destination location (e.g. field): 5
Number of pipes: 9
Pipe 1 -> from to capacity(litres/min): 0 1 16
Pipe 2 -> from to capacity(litres/min): 0 2 13
Pipe 3 -> from to capacity(litres/min): 1 3 12
Pipe 4 -> from to capacity(litres/min): 2 1 4
Pipe 5 -> from to capacity(litres/min): 2 4 14
Pipe 6 -> from to capacity(litres/min): 3 2 9
Pipe 7 -> from to capacity(litres/min): 3 5 20
Pipe 8 -> from to capacity(litres/min): 4 3 7
Pipe 9 -> from to capacity(litres/min): 4 5 4
Maximum Flow: 23 litres/minute
```

```
Enter your choice: 5

========== CROP -> FIELD ASSIGNMENT (Bipartite Matching) ==========
Number of fields available: 4
Fields (1..4) suitable for Rice (space separated): 1 2
Fields (1..4) suitable for Wheat (space separated): 1
Fields (1..4) suitable for Maize (space separated): 1 3
Fields (1..4) suitable for Cotton (space separated): 2 4
Fields (1..4) suitable for Groundnut (space separated): 3 4
Fields (1..4) suitable for Sugarcane (space separated): 4
Fields (1..4) suitable for Tomato (space separated):
Fields (1..4) suitable for Chilli (space separated):

Crops that can be planted at the same time: 4
  Rice -> Field 1
  Wheat -> not assigned
  Maize -> Field 3
  Cotton -> Field 2
  Groundnut -> Field 4
  Sugarcane -> not assigned
  Tomato -> not assigned
  Chilli -> not assigned
```

Fig. 5.8 Crop-to-field assignment result (Bipartite Matching)

STEP-6

Command Prompt - java Main

```
Enter your choice: 6

========== FIELD VISIT ROUTE PLANNER (Bitmask DP / TSP) ==========
Number of locations (location 0 = farm, max 15): 5
Distances from location 0 to 0..4: 0 10 15 20 25
Distances from location 1 to 0..4: 10 0 35 25 30
Distances from location 2 to 0..4: 15 35 0 30 20
Distances from location 3 to 0..4: 20 25 30 0 15
Distances from location 4 to 0..4: 25 30 20 15 0
Shortest round trip: 0 -> 2 -> 4 -> 3 -> 1 -> 0
Total distance: 85
```

Fig. 5.9 Field visit route (Bitmask DP)

STEP-7

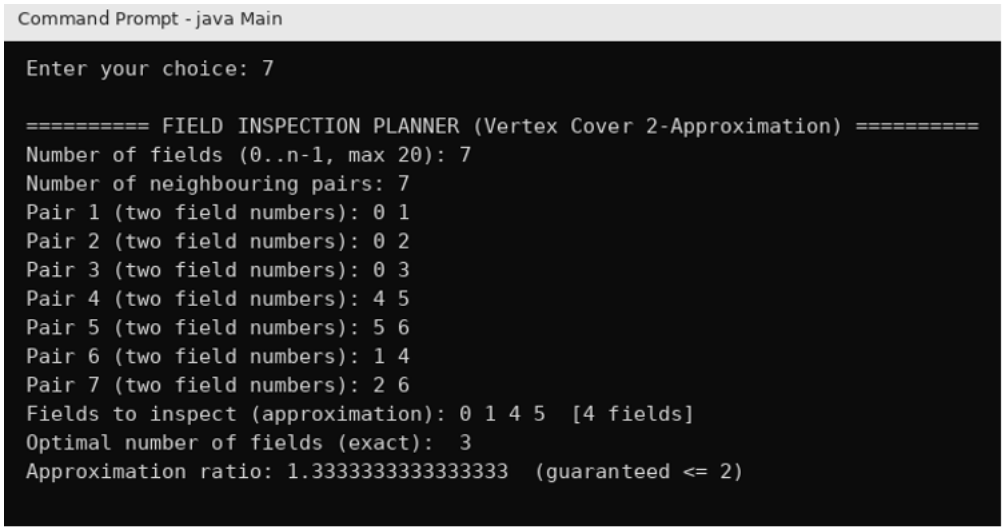


Fig. 5.10 Field inspection planner (Vertex Cover 2-approximation)